

4-Bit Binary Adder - Arduino Implementation

Header

Project Name: 4-Bit Binary Adder

Version: 1.0

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Programming Language: C++

Project Overview

This project implements a 4-bit binary adder using an Arduino microcontroller. The system reads two 4-bit binary numbers from input pins, performs binary addition, and displays the result (4-bit sum and carry bit) on output pins. This project demonstrates fundamental digital logic concepts and Arduino programming skills.

Project Structure

```
4-bit-binary-adder/
├── src/
│   └── 4_bit_adder.ino  # Main Arduino sketch
├── docs/
│   └── circuit_diagram.png # Wiring diagram (to be added)
├── README.md           # Project documentation
└── LICENSE              # Project license
```

Components Used

- Arduino Uno or compatible board
- 8x toggle switches or digital input sources (for the two 4-bit numbers)
- 5x LEDs (for the 4-bit result display)
- 1x LED (for the carry bit indicator)
- Breadboard and connecting wires
- Resistors (220-330Ω for LEDs)

Circuit Connection Details

Input Pins (4-bit numbers):

- First Number (A):
 - Bit 0 (LSB): Arduino Pin 0
 - Bit 1: Arduino Pin 1
 - Bit 2: Arduino Pin 2
 - Bit 3 (MSB): Arduino Pin 3
- Second Number (B):
 - Bit 0 (LSB): Arduino Pin 4
 - Bit 1: Arduino Pin 5
 - Bit 2: Arduino Pin 6
 - Bit 3 (MSB): Arduino Pin 7

Output Pins (Result):

- Sum Output:
 - Bit 0 (LSB): Arduino Pin 8
 - Bit 1: Arduino Pin 9
 - Bit 2: Arduino Pin 10
 - Bit 3 (MSB): Arduino Pin 11
- Carry Bit: Arduino Pin 12

How It Works

Binary Addition Logic

The project implements a 4-bit binary adder using a series of full adders:

- Each full adder takes three inputs (two bits from input numbers and a carry-in)
- Produces two outputs (sum bit and carry-out)
- The carry-out from each stage becomes the carry-in for the next higher bit

Code Implementation

The Arduino sketch includes:

1. Setup Function: Configures input and output pins
2. Loop Function: Continuously reads inputs, performs addition, and updates outputs
3. Helper Functions:
 - `sumBit()`: Calculates the sum bit using XOR logic
 - `sumCarryBit()`: Calculates the carry bit using AND-OR logic

Operational Flow

1. Read two 4-bit numbers from input pins
2. Initialize carry bit to 0
3. Process each bit from least significant to most significant:
 - Calculate sum bit for current position
 - Calculate carry bit for next position
4. Display the 4-bit result and final carry bit on output pins

Getting Started

Prerequisites

- Arduino IDE installed on your computer
- Basic understanding of binary arithmetic and digital logic
- Familiarity with Arduino programming

Installation

1. Clone or download this repository
2. Open the `4_bit_adder.ino` file in Arduino IDE
3. Connect your Arduino board to the computer
4. Wire the circuit according to the connection details
5. Upload the sketch to your Arduino

Usage

1. Set the input pins (0-7) to HIGH (5V) or LOW (0V) to represent binary 1 or 0
2. The output pins (8-12) will light up LEDs to show the binary result
3. Pin 12 indicates the carry bit (overflow)

Applications

- Educational tool for learning binary arithmetic
- Foundation for more complex arithmetic logic units (ALUs)
- Demonstrates fundamental digital logic concepts
- Can be expanded to handle subtraction and other operations

Future Enhancements

- Add support for binary subtraction

- Implement overflow detection
- Add seven-segment display output for decimal representation
- Create a library for reusable binary operations

License

This project is open source and available under the MIT License.

Contributing

Contributions are welcome! Please feel free to submit issues, suggestions, or pull requests to improve this project.

Acknowledgments

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- Instructor: Glauber Kiss de Souza
- Course: Technically Oriented Analysis